Enhanced Langlands parameters of regular principal series

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Let G be a connected reductive group over a finite extension $F/\mathbb{Q}_p$ with residue field of order q_F.

The local Langlands correspondence

$$\left\{\begin{array}{c}\text{irreducible smooth}\\ \text{representations}\\ \pi \text{ of } G(F)\end{array}\right\}/\text{iso.} \xrightarrow{\mathcal{L}_G} \left\{\begin{array}{c}L\text{-parameters for } G\\ \phi : W_F \times \mathrm{SL}_2(\mathbb{C}) \to {}^L G\end{array}\right\}/G^\vee\text{-conj.}$$

Write $\Phi(G)$ for L-parameters. Each fibre is a finite L-packet.

$$\Pi_\phi = \mathcal{L}_G^{-1}([\phi]), \qquad [\phi] \in \Phi(G).$$

How can we refine the local Langlands correspondence to obtain a bijection?

Several representations can share the same parameter ϕ. The enhanced LLC seeks a bijection by labelling each packet member on the Galois side.

$$\pi \quad \longleftrightarrow \quad (\phi_\pi, \rho_\pi).$$

The **enhancement** ρ_π labels π within its packet. It is an irreducible representation of a finite component group attached to ϕ_π.

The component group for quasi-split G

Let $C_\phi = Z_{G^\vee}(\operatorname{im}\phi)$ be the centralizer of the parameter. Then

$$\mathcal{R}_\phi = \pi_0\Big(C_\phi / Z(G^\vee)^{W_F}\Big), \qquad \rho_\pi \in \operatorname{Irr}(\mathcal{R}_\phi) \quad (\pi \in \Pi_\phi).$$

Here π_0 denotes the group of connected components.

The category of smooth complex representations has the **Bernstein decomposition**

$$\mathrm{Rep}(G(F)) \simeq \prod_{\mathfrak{s}\in\mathfrak{B}(G)} \mathrm{Rep}(G(F))^{\mathfrak{s}}.$$

Every representation decomposes canonically as $\pi \simeq \bigoplus_{\mathfrak{s}} \pi^{\mathfrak{s}}$.

Here $\mathfrak{B}(G)$ consists of inertial classes $\mathfrak{s} = [M, \sigma]_G$, where M is a Levi subgroup and σ is supercuspidal. Inertial equivalence allows conjugacy and unramified twists of σ.

The block $\mathrm{Rep}(G(F))^{\mathfrak{s}}$ consists of representations whose irreducible subquotients have supercuspidal support in $\mathfrak{s}$. For principal series, $M = T$ and $\sigma = \chi$ is a torus character.

Example for $G = \mathrm{GL}_2$

$$\mathbf{1}, \mathrm{St} \in \mathrm{Irr}(G(F))^{[T,\mathbf{1}]_G}, \qquad \phi_{\mathbf{1}} \not\sim \phi_{\mathrm{St}}.$$

Representations in one Bernstein block can belong to different L-packets.

Solleveld (2025)

For quasi-split G with a fixed Whittaker datum, the principal-series LLC is a bijection.

$$\left\{\begin{array}{c}\text{irreducible smooth}\\ \text{principal-series}\\ \text{representations of } G(F)\end{array}\right\}/\text{iso.} \xrightarrow[\sim]{\mathcal{L}^e_{\mathrm{ps}}} \left\{\begin{array}{c}\text{principal-series enhanced}\\ L\text{-parameters } (\phi,\rho)\\ \rho \in \mathrm{Irr}(\mathcal{R}_\phi)\end{array}\right\}/G^\vee\text{-conj.}$$

Forgetting the enhancement

$$\pi \longmapsto (\phi_\pi, \rho_\pi) \longmapsto \phi_\pi.$$

Projection onto the first entry recovers the ordinary packet map.

A trivial enhancement $\rho = 1$ need not mean a trivial component group.

Let ${}^L G = G^\vee \rtimes W_F$. An L-**parameter** is an admissible homomorphism

$$\phi : W_F \times \mathrm{SL}_2(\mathbb{C}) \longrightarrow {}^L G, \qquad \text{up to } G^\vee\text{-conjugacy.}$$

It is continuous on W_F and algebraic on $\mathrm{SL}_2(\mathbb{C})$.

Equivalent Weil–Deligne form

$$(\widetilde{\phi}, N), \qquad \widetilde{\phi} : W_F \longrightarrow {}^L G, \qquad N \in \mathrm{Lie}(G^\vee) \text{ nilpotent},$$
$$\widetilde{\phi}(w) := \phi\Big(w, \mathrm{diag}(\|w\|^{1/2}, \|w\|^{-1/2})\Big).$$
$$\mathrm{Ad}(\widetilde{\phi}(w))N = \|w\| N.$$

An **enhanced** L-**parameter** can therefore be written as a triple $(\widetilde{\phi}, N, \rho)$. Here $\widetilde{\phi}$ is the **infinitesimal parameter**, N the **monodromy**, and ρ the **enhancement**.

The setting: regular principal series

Let G/F be connected, quasi-split, and reductive, with an F-Borel $B = TU$. For a smooth character $\chi : T(F) \to \mathbb{C}^\times$, form

$$I(\chi) = \mathrm{Ind}_B^G \chi.$$

The character χ is **regular** if

$$\mathrm{Stab}_W(\chi) = \{1\}.$$

Here $W = N_G(S)(F)/T(F)$ is the relative Weyl group, where $S \subset T$ is maximal F-split.

- The torus LLC determines the common infinitesimal parameter of the constituents. It attaches $\widehat{\chi} : W_F \to {}^L T \hookrightarrow {}^L G$ to χ, and

$$[\widetilde{\phi}_\pi] = [\widehat{\chi}] \quad \text{for every irreducible constituent } \pi \text{ of } I(\chi).$$

- We want the monodromy and enhancement of *each* irreducible constituent of $I(\chi)$.

Every constituent of $I(\chi)$ has the common infinitesimal parameter

$$\widetilde{\phi} = \widehat{\chi}.$$

We seek its monodromy N and enhancement ρ.

For regular χ, the unique irreducible subrepresentation is $\operatorname{soc} I(\chi)$.

The motivating question

When is $N_{\mathrm{soc}} = 0$? Can this be decided from reducibility over F?

Bellaïche–Chenevier (2009) gave a sufficient condition after passing to a splitting field. We seek an intrinsic necessary and sufficient condition over F.

Earlier results on regular principal series

- **Rodier (1981), split groups.**
  Classified the irreducible constituents by chambers of the reducibility arrangement.
- **Rodier (1984), split groups.**
  Described their L-parameters, including a zero-monodromy criterion for the socle.
- **Luo (2021), general reductive groups.**
  Extended the chamber classification to regular generalized principal series, with Jacquet modules, multiplicity one, and independent reducibility coroots.
- **Solleveld (2025).**
  His enhanced principal-series LLC and joint work on geometric Hecke algebras provide a framework in which the enhancement can be computed.

Our aim: compute the monodromy N and enhancement ρ of every constituent of regular principal series for arbitrary quasi-split groups.

Reducibility directions and oriented walls

For each relative root line, test the rank-one induction

$$\mathrm{Ind}_{B\cap M_\alpha}^{M_\alpha}(\chi).$$

On a reducible root line, choose an integral coroot map $H_a : F^\times \to S(F) \subset T(F)$. Choose the sign of a so that

$$|\chi(H_a(t))| = |t|^{s_a}, \qquad s_a > 0.$$

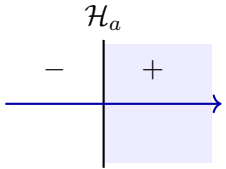
Collect these directions in $\mathcal{S}(\chi)$. Set

$$\mathfrak{a}_T^* := \mathrm{Hom}\big(T(F)/T(F)_{\mathrm{cpt}}, \mathbb{R}\big),$$

where $T(F)_{\mathrm{cpt}}$ is the maximal compact subgroup of $T(F)$.

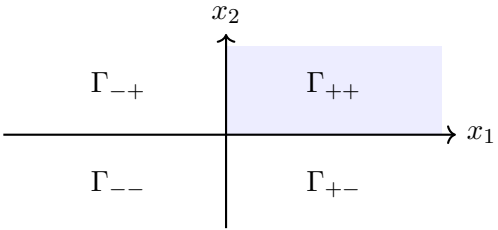
$$\mathcal{H}_a = \{x \in \mathfrak{a}_T^* : \langle x, a\rangle = 0\},$$
$$\mathcal{H}_a^+ = \{x \in \mathfrak{a}_T^* : \langle x, a\rangle > 0\}.$$

The connected regions are the chambers.



Rodier–Luo

The directions in $\mathcal{S}(\chi)$ are linear independent. Chambers label the pairwise nonisomorphic constituents of $I(\chi)$, each with multiplicity one.



Use $x_i = \langle x, a_i \rangle$. Two independent directions give four sign patterns and four constituents. In general: $2^{|\mathcal{S}(\chi)|}$ constituents.

Main theorem: positive sides determine monodromy, with trivial enhancement

Each reducibility direction $a \in \mathcal{S}(\chi)$ determines a one-dimensional vector space. Choose a nonzero vector $X_{-\iota(a)}$ in that space.

Li–Yang–Zhang–Zhao

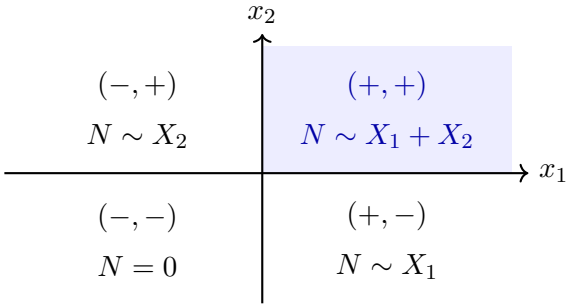
The enhanced parameter of π_Γ has a representative with

$$\widetilde{\phi}_\Gamma = \widehat{\chi}, \qquad N_\Gamma \sim \sum_{\substack{a \in \mathcal{S}(\chi) \\ \Gamma \subset \mathcal{H}_a^+}} X_{-\iota(a)}, \qquad \rho_\Gamma = \mathbf{1}.$$

Here $\sim$ denotes conjugacy under $Z_{G^\vee}(\widehat{\chi}(W_F))^\circ$.

- All irreducible constituents of the principal series $I(\chi)$ share the same infinitesimal parameter.
- The chamber determines the monodromy support.
- Every enhancement is trivial; the component group need not be.

Reading the formula: four chambers, four supports



Here $X_i = X_{-\iota(a_i)}$. All four constituents share $\widetilde{\phi} = \widehat{\chi}$. All enhancements are trivial, even though the component groups may differ.

A positive sign includes that root direction in the support.

When does the socle have zero monodromy?

Let C^+ be the dominant Weyl chamber determined by B, and let Γ_{soc} be the arrangement chamber containing it.

$$\operatorname{soc} I(\chi) = \pi_{\Gamma_{\mathrm{soc}}}.$$

A criterion using reducibility over F

$$N_{\mathrm{soc}} = 0 \quad \Longleftrightarrow \quad \langle x, a \rangle < 0 \quad (\forall x \in C^+,\ \forall a \in \mathcal{S}(\chi)).$$

Equivalently, C^+ lies on the negative side of every oriented reducibility wall. In this case,

$$\phi_{\mathrm{soc}}(w, g) = \widehat{\chi}(w), \qquad \rho_{\mathrm{soc}} = \mathbf{1}.$$

GL_2: two inducing orders

Let $\nu = |\cdot|_F$, let B be upper triangular, and put $s = \mathrm{diag}(q_F^{-1/2}, q_F^{1/2})$.

$$0 \longrightarrow \mathrm{St} \longrightarrow \mathrm{Ind}_B^{\mathrm{GL}_2}(\nu^{1/2} \boxtimes \nu^{-1/2}) \longrightarrow \mathbf{1} \longrightarrow 0.$$

Constituent	Wall side	$(\widetilde{\phi}(\mathrm{Fr}_F), N)$	ρ
St (socle)	$+$	(s, E_{12})	$\mathbf{1}$
$\mathbf{1}$ (quotient)	$-$	$(s, 0)$	$\mathbf{1}$

$$0 \longrightarrow \mathbf{1} \longrightarrow \mathrm{Ind}_B^{\mathrm{GL}_2}(\nu^{-1/2} \boxtimes \nu^{1/2}) \longrightarrow \mathrm{St} \longrightarrow 0.$$

Constituent	Wall side	$(\widetilde{\phi}(\mathrm{Fr}_F), N)$	ρ
$\mathbf{1}$ (socle)	$-$	$(s^{-1}, 0)$	$\mathbf{1}$
St (quotient)	$+$	(s^{-1}, E_{21})	$\mathbf{1}$

In each order, the constituents share an infinitesimal character but belong to different L-packets.

The socle in the lower table is $\mathbf{1}$, with zero monodromy.

Proof roadmap

$$\underbrace{\pi_\Gamma}_{\text{representation}} \mapsto \underbrace{M_\Gamma}_{\substack{\text{affine Hecke}\\ \text{module}\\ \text{(representation side)}}} \mapsto \underbrace{M_\Gamma^\vee}_{\substack{\text{affine Hecke}\\ \text{module}\\ \text{(Galois side)}}} \mapsto \underbrace{V_\Gamma}_{\substack{\text{graded Hecke}\\ \text{module}}} \mapsto \underbrace{(\sigma_\Gamma, N_\Gamma, \rho = \mathbf{1})}_{\substack{\text{geometric}\\ \text{parameter}}} \mapsto \underbrace{(\widetilde{\phi}_\Gamma, N_\Gamma, \rho_\Gamma)}_{\substack{\text{enhanced}\\ L\text{-parameter}}}.$$

1. **Bernstein block: pass to an affine Hecke module.**
2. **Hecke algebra comparison: transfer to the Galois side.**
3. **Fix the central character and reduce to a graded Hecke module.**
4. **Geometry: identify the triple attached to V_Γ.**
5. **Translate the triple into the enhanced Langlands parameter.**

1. **The space of possible monodromy.** Fix the semisimple datum. For $a \in \mathcal{S}(\chi)$, put $\alpha_a := -v^{-1}\iota(a)$. Then

$$\text{Possible Monodromy} = \bigoplus_{a \in \mathcal{S}(\chi)} \mathbb{C} X_{\alpha_a}.$$

   Every element of this space is nilpotent.

2. **Conjugacy classes are parametrized by subsets.** Independent root characters allow the centralizing torus to rescale nonzero coordinates:

$$\{\text{possible monodromy conjugacy classes}\} \longleftrightarrow \{J \subseteq \mathcal{S}(\chi)\}, \qquad N_J = \sum_{a \in J} X_{\alpha_a}.$$

3. **Which subset corresponds to the representation π_Γ?**

Proof: the chamber determines monodromy

Use weights to determine the subset: Let σ_Γ be a selected weight of V_Γ.

$$\pi_\Gamma \longmapsto M_\Gamma \longmapsto M_\Gamma^\vee \longmapsto V_\Gamma, \qquad \dim(V_\Gamma)_{\sigma_\Gamma} = 1.$$

1. **Geometric classification.** For an eligible triple $(\sigma_\Gamma, N_J, \rho)$, the standard module E_J has a unique irreducible quotient L_J.
2. **Weights and Möbius inversion.** Weight calculations and Möbius inversion determine the subset

$$J_\Gamma = \{a \in \mathcal{S}(\chi) : \Gamma \subset \mathcal{H}_a^+\}.$$

In particular,

$$\boxed{V_\Gamma \simeq L_{J_\Gamma}, \qquad N_\Gamma \sim \sum_{a \in J_\Gamma} X_{\alpha_a}.}$$

Proof: why the enhancement is trivial

Use generalized Springer theory for the monodromy N_Γ.

1. **Possible enhancements.** A possible enhancement $\rho \in \mathrm{Irr}(\pi_0 Z_H(\sigma_\Gamma, N_\Gamma))$ must occur in $H_*(\mathcal{P}_{N_\Gamma}^{\sigma_\Gamma}; \mathbb{C})$ after specialization.
2. **Only trivial enhancements appear.** Regularity makes the fixed locus $\mathcal{P}_{N_\Gamma}^{\sigma_\Gamma}$ finite and discrete. The group $\pi_0 Z_H(\sigma_\Gamma, N_\Gamma)$ acts trivially on its homology.

Thus the possible enhancement for the triple $(\sigma_\Gamma, N_\Gamma, \rho)$ is trivial:

$$\boxed{\rho = \mathbf{1}.}$$

The component group itself may still be nontrivial.

Hecke modules and enhanced L-parameters

At the chosen bounded datum and central character, the parametrizations complete the chain:

$$\pi_\Gamma \longmapsto M_\Gamma \longmapsto M_\Gamma^\vee \longmapsto V_\Gamma \longmapsto (\sigma_\Gamma, N_\Gamma, \rho = \mathbf{1}) \longmapsto (\widetilde{\phi}_\Gamma, N_\Gamma, \rho_\Gamma).$$

- **Infinitesimal character.** In the chosen coordinates, $\widetilde{\phi}_\Gamma = \widehat{\chi^v}$.
- **Monodromy.** The nilpotent entry N_Γ is the parameter's monodromy operator.
- **Enhancement.** The representation ρ_Γ is determined by inflation:

$$\rho = q^* \rho_\Gamma, \qquad q : \pi_0 Z_H(\sigma_\Gamma, N_\Gamma) \twoheadrightarrow \mathcal{R}_{\phi_\Gamma}.$$

References. AMS, *Affine Hecke algebras for Langlands parameters*;
Solleveld, *On principal series representations of quasi-split reductive p-adic groups*.

What the formula adds to the LLC dictionary

1. **An explicit constituent-by-constituent answer.**
   The inducing character fixes $\widetilde{\phi}$; chamber signs determine N; every enhancement is trivial.

2. **A criterion over the original field.**
   The socle has zero monodromy exactly when its chamber lies on every negative side.

3. **A uniform quasi-split formula.**
   It combines known chamber labels with the established principal-series LLC, including nonsplit and ramified cases.

Thank you.